

Chapter 3. **GROUPS**

3.4 Homomorphisms and isomorphisms



Today we will

- learn some properties of group homomorphisms

THEOREM 3.38. Let $\varphi: G_1 \rightarrow G_2$ be a group homomorphism. Then:

- (1) φ preserves the identity element: $\varphi(1_{G_1}) = 1_{G_2}$
- (2) φ preserves inverses: $\varphi(g^{-1}) = \varphi(g)^{-1}$ for all $g \in G_1$.
- (3) φ preserves powers: $\varphi(g^k) = \varphi(g)^k$ for all $g \in G_1$ and $k \in \mathbb{Z}$.

Proof.

$$\textcircled{1} \quad \varphi(1_{G_1}) \stackrel{?}{=} 1_{G_2}$$

Notice that

$$\varphi(1_{G_1}) = \varphi(1_{G_1} \cdot 1_{G_1}) \stackrel{\substack{\varphi \text{ is a group} \\ \text{homomorphism}}}{=} \varphi(1_{G_1}) \varphi(1_{G_1}).$$

Using the Cancellation Law in G_2 :

$$\underline{1_{G_2} = \varphi(1_{G_1})}.$$

(2) For $g \in G_1$, we want to show
that $\varphi(g^{-1}) \stackrel{?}{=} \varphi(g)^{-1}$.

$$1_{G_2} \stackrel{①}{=} \varphi(1_{G_1}) = \varphi(gg^{-1}) \overset{\substack{\varphi \text{ is a group} \\ \text{homomorphism}}}{=} \varphi(g)\varphi(g^{-1}).$$

Similarly, we have that

$$\varphi(g^{-1})\varphi(g) = 1_{G_2}.$$

This shows that $\varphi(g)^{-1} = \varphi(g^{-1})$.

$$\textcircled{3} \quad \varphi(g^k) \stackrel{?}{=} \varphi(g)^k \text{ for all } g \in G, \text{ and } \underline{k \in \mathbb{Z}}$$

We first use induction to show that

$$\varphi(g^n) = \varphi(g)^n \text{ for all } n \geq 0.$$

For $n=0$

$$\varphi(g^0) = \varphi(1_{G_1}) \stackrel{\textcircled{1}}{=} 1_{G_2} = \varphi(g)^0 \quad \checkmark$$

Suppose that $\varphi(g^k) = \varphi(g)^k$ and let us show that $\varphi(g^{k+1}) = \varphi(g)^{k+1}$.

$$\begin{aligned} \underline{\varphi(g^{k+1})} &\stackrel{\text{def.}}{=} \varphi(g^k \cdot g) \underset{\substack{\varphi \text{ group} \\ \text{hom}}}{=} \varphi(g^k) \varphi(g) \stackrel{\text{I.H.}}{=} \\ &= \varphi(g)^k \varphi(g) \stackrel{\text{def.}}{=} \underline{\varphi(g)^{k+1}} \end{aligned}$$

If $k < 0$, then $n := -k > 0$.

$$\underline{\varphi(g^k)} = \varphi(g^{-n}) \stackrel{\text{def.}}{=} \varphi((g^n)^{-1}) \stackrel{\textcircled{2}}{=} \underline{\varphi(g^n)^{-1}}$$

$$\begin{aligned}
 &= \left(\varphi(g)^m \right)^{-1} \\
 &\stackrel{\text{def}}{=} \varphi(g)^{-m} = \varphi(g)^k
 \end{aligned}$$

③ is true for all positive numbers and 0



THEOREM 3.39. Let G , G_1 and G_2 denote groups.

- (1) The identity map is an isomorphism for every group.
- (2) If $\varphi: G \rightarrow G_1$ is an isomorphism, then the inverse $\varphi^{-1}: G_1 \rightarrow G$ is also an isomorphism.
- (3) If $\varphi: G \rightarrow G_1$ and $\psi: G_1 \rightarrow G_2$ are isomorphisms, then $\psi \circ \varphi: G \rightarrow G_2$ is also an isomorphism.

COROLLARY 3.40. The isomorphic relation $\cong$ is an equivalence relation.

COROLLARY 3.41. Then the set $\text{aut}(G)$ consisting of all automorphisms of a group G forms a group under composition. It is called the **automorphism group** of G .

Proof

② Suppose that $\varphi: G \rightarrow G_1$ is an isomorphism.

We want to prove that $\varphi^{-1}: G_1 \rightarrow G$ is also an isomorphism. We only need to check that

$$\varphi^{-1}(a_1 b_1) \stackrel{?}{=} \varphi^{-1}(a_1) \varphi^{-1}(b_1) \text{ for}$$

all $a_1, b_1 \in G_1$.

There exist $a, b \in G$ such that

$\psi(a) = a_1$ and $\psi(b) = b_1$, (ψ is surjective),

ψ is a group hom.

$$\psi^{-1}(a_1, b_1) = \psi^{-1}(\psi(a)\psi(b)) \stackrel{\psi \text{ is a group hom.}}{=} \psi^{-1}(\psi(ab))$$

$$\stackrel{\psi^{-1}\psi = \text{Id}}{=} ab = \psi^{-1}(\psi(a))\psi^{-1}(\psi(b))$$

$$= \psi^{-1}(a_1)\psi^{-1}(b_1)$$

③ Suppose now that $\psi: G \rightarrow G_1$ and $\varphi: G_1 \rightarrow G_2$ are isomorphisms. We want to prove that

$$\varphi \circ \psi: G \rightarrow G_2$$

is an isomorphism.

We have that $\varphi \circ \varphi$ is bijective
(composition of bijective maps is bijective).
So, it remains to show that

$$\underline{(\varphi \circ \varphi)(ab) \stackrel{?}{=} (\varphi \circ \varphi)(a)(\varphi \circ \varphi)(b)}, \text{ for}$$

all $a, b \in G$.

Take $a, b \in G$.

$$\begin{aligned} (\varphi \circ \varphi)(ab) &= \varphi(\varphi(ab)) \stackrel{\varphi \text{ group hom.}}{=} \\ &= \varphi(\varphi(a)\varphi(b)) \stackrel{\varphi \text{ group hom.}}{=} \\ &= \varphi(\varphi(a)) \varphi(\varphi(b)) \\ &= (\varphi \circ \varphi)(a)(\varphi \circ \varphi)(b) \end{aligned}$$

